Science · LAUNCH · Autumn 2 · Week 2 · SCI-LAUNCH-A2-W02 · Date _____

Growth and stem cells

Today I can… ☐ I can distinguish making more cells from cells taking on a specialised job. ☐ I can compare animal growth with plant growth. ☐ I can identify a role for an animal stem cell or a plant meristem.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Staff answers · retrieval and model

Starter

1. Two daughter body cells with the same chromosome set. Body-cell mitosis followed by cell splitting makes the two cells.

2. Active transport.

3. The nucleus. Confirm the actual taught coverage before using retrieval.

I do check

No. Making more cells does not by itself give them a specialised tissue job; differentiation gives specialised roles.

Full model

In skin, mitosis adds cells; some stem-cell descendants differentiate for a tissue job. At a shoot tip, meristem cells divide, then elongate and differentiate.

We do matches

Makes more cells: division. Plant cells become longer: elongation. Cell gains a tissue job: differentiation. Keep the source clue and choice order.

Missing step

Elongation lengthens plant cells. Division and differentiation occur in both example routes; elongation is an additional plant-growth step in this model.

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Staff answers · hinge and routes

Hinge answer A

The gut-lining and nerve cells differentiated into specialised cells with different jobs despite sharing inherited DNA. B is false: differentiation does not require loss of every inherited gene. C confuses shared DNA with identical cell jobs. D wrongly rules out prior cell division.

Decision for the pair

Collect both learners' responses together. Proceed when both show a secure idea. If either is unsure, re-model a dividing stem cell and a specialised skin-cell branch for that learner, then ask the parallel hinge. The other learner can continue. If both need the model, hear them separately afterwards.

Parallel hinge

A muscle cell and a liver cell share inherited DNA. Must their jobs be identical? No. Differentiation produces specialised cell types.

Independent F and M

F: Skin branch uses divide then differentiate. Plant branch uses divide, elongate and differentiate. M: division makes more cells; differentiation gives a specialised job.

Independent H

Both growth examples involve division and differentiation. The plant shoot also grows through elongation. Animal stem-cell descendants may specialise; plant meristem cells divide at growing tips.

Go deeper and repair audit

More cells without new cell types shows division, but not differentiation into new types. Two defensible repairs: add the differentiation step after division; relabel a specialised descendant by its tissue job instead of calling every new skin cell a stem cell. Accept two justified corrections consistent with the source model.

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Staff answers · adaptations and access

Teaching sequence and fade

Model the complete skin/shoot comparison. In We do, remove only the plant elongation label. In Independent, give no filled answer. For a later fade, hide the animal differentiation arrow as well and ask for both missing steps.

Designed for two learners

Quiet pair rehearsal supports the same comparison. Hear each learner's own reason and retain separate evidence. Agree adult roles before the lesson. With two adults, one can support a learner while the other hears the partner, then swap. Follow the actual staffing plan when one adult attends.

Planned adaptations

Cognition: use a high-contrast two-branch strip and remove only one cue. Communication: accept point, sign, speech or an agreed communication aid. SEMH: privately preview the comparison, agree a pause signal and return to one card. Sensory/physical: large easy-grip cards or an accessible response route. Language: show four science words with pictures and examples; allow planning in any language and keep M/H challenge open.

Regulation

Check in before the hinge and Independent. At the agreed pause signal, give space and one return step. Re-enter at the last completed step. Do not gather evidence while distressed.

Method and limits

The cards are prepared model rehearsal, not a claimed core practical. Use the desk mat and pick up dropped cards. Confirm the school risk assessment and exact kit. The skin-repair scene is illustrative, not evidence of an individual treatment outcome.

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Staff answers · feedback and evaluation

Feedback and evidence

VF when feedback happens; WS with actual prompt level; I when independent; NS(+) as a precise next step; // for a true self-edit; ? for a requested read-back. R follows a pupil response to the next step and adult judgement, never automatic. E requires an actual observable curriculum action.

Record separately

Record each learner's hinge choice, explanation, named misconception, valid response mode and deepest prompt. Label book or captured evidence SCI-LAUNCH-A2-W02. Keep pupil data out of the page.

Assessment references

[AWARD] [QUAL] topic checks and end-of-unit evidence are masked as required. Record only the actual curriculum action, with its lesson code.

If behind

Preserve the independent comparison of division and cell jobs. After a secure hinge, shorten repeated oral rehearsal. If the exit reflection moves to the next Arrival, keep core evidence and safety, record the change and pass on any unresolved misconception. Do not replace incomplete individual work with a shared product.

Individual targets

Record the learner's own sequencing/independence target, communication mode, private regulation/re-entry target and access choices. Use the actual observed response and support level, not a diagnosis. Planning in a first language keeps the science demand open.

Evaluation

Date taught: _____ Delivered as planned: □ Yes □ Partly □ No

Criteria met: □ All □ Most □ Some

Adaptations and coverage impact: _____

Next model or prompt from the Exit response: _____

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